

CALIPER

An enrichment pipeline that measures what it claims,
and refuses to write what it cannot support.

1,000

ROWS IN 9S

252

COLUMNS OUT

26,627

TRACEABLE CELLS

42

TESTS PASSING

Six columns in. Two hundred and fifty-two out.

One real row from the working dataset. Three of its five fields are placeholders that mean “empty”.

```
Mfg_Part_Num : 49-94-1940
Part_Desc    : 49-94-1940 Milw 14"x1/8"x1"
              Masonry Cut Off Disc
El_Brand     : -- Unbranded --
DIB_Brand    : -- No DIB Brand --
Part_Manuf   : Milwaukee Accessory (4031)
```

1

An approved brand

exact casing, with the ® symbol

2

A classpath

the key every attribute is validated against

3

Five descriptions

five different length and casing rules

4

Ordered attributes

drawn from a controlled vocabulary

The brief is blunt about the obvious approach: “a fluent description made of invented values scores zero.”

Facts before prose.

Every extractor writes into one structure. Nothing downstream may invent a value — it can only render facts that already exist.

brand	Milwaukee®	IDN-BRD-01	0.88
manufacturer	Milwaukee Tool	IDN-MFR-01	0.86
dimensions	14 in x 1/8 in x 1 in	DIM-CHN-01	0.92
diameter	14 in	DIM-CHN-02	0.92
thickness	1/8 in	DIM-CHN-02	0.92
arbor_size	1 in	DIM-CHN-02	0.92
application	Masonry	APP-MAT-01	0.92
item_type	Cut Off Disc	ITM-LEX-01	0.95

→ Milwaukee® 49-94-1940 Cut Off Disc,
14 in x 1/8 in x 1 in, Masonry

1

Nothing ungrounded gets in

A fact without evidence is rejected at insertion. A tested invariant, not a convention.

2

The five descriptions agree

They are five renderings of one fact set, not five separate generations.

3

Provenance is free

26,627 cells, each traceable to the rule that made it and the characters behind it.

Rules, registries, taxonomy, family consensus and the model all write facts. Composition, validation and export only read them.

Four findings that came from measuring, not reasoning.

Each one changed the build. Two contradicted decisions already made.

01 The answer key fills 25% of the sheet

UPC, UNSPSC, every dimension, SDS and country of origin are blank in both published rows. A pipeline filling 90% is inventing two-thirds of a catalogue.

02 Attribute values are not in the supplier row

The labelled dishwasher's input is 39 characters. Eleven of its twelve attribute values appear nowhere in it — they need the manufacturer's source.

03 A guardrail caught our own parser

Four wheels reported an arbor hole wider than the disc. The dimension regex knew inches but not millimetres. Findings went $4 \rightarrow 2 \rightarrow 0$; both defects were real.

04 A bug no reading could find

A heredoc turned `\b` into a literal backspace byte inside two regexes. `grep`, `sed` and `cat` all showed the files as correct — terminals don't draw it.

RESULTS

1,000 rows, nine seconds, no API key required.

100%

INVOICE $\leq$ 40 CHARS

compliant by construction, not by checking

92.5%

BRAND RESOLVED

to an approved name with its ® symbol

88.8%

CLASSIFIED

the remainder abstain rather than guess

77.4%

READY TO PUBLISH

774 rows need no human at all

99.5%

SIBLING AGREEMENT

over 1,516 comparisons, 185 families

614

RELATIONSHIP EDGES

powers · variant_of · same_series

64

CATEGORY SPECS

induced from rows, not hand-written

48.0%

VS. THE ANSWER KEY

on 2 labelled rows — a narrow base, stated

Two labelled rows is a narrow base. Consistency is not.

No care widens a calibration set of two — but self-consistency can be asked of all 1,000 rows: products in one family must agree about the facts they share.

```
families with 2+ members : 185
attribute comparisons    : 1,516
siblings agree           : 99.5%

brand 100%      manufacturer 100%
item_type 99.5% classpath 99.4%
```

It earned its place within the hour.

Brand agreement came back at 95.3%, which reads as an extraction error. It was a clustering error — every “Dishwasher SS” from one appliance co-op landed in a single family, because the brand lives in the part number, not the description.

Making family membership brand-aware took brand and manufacturer to 100%.

This is deliberately not called accuracy — a pipeline can be uniformly wrong and perfectly consistent. But an extractor that disagrees with itself across products differing only by size is certainly wrong somewhere.

Upload a catalogue. Click any cell. See why.

A judge's own file works — column names are detected, not assumed. Proven on a test file with different headers, different order and two junk columns.

Evidence panel

Every value names the rule that produced it, the characters of the input that justified it, and its confidence.

Budget solver trace

The 40-character invoice line, showing which facts were kept, which abbreviated, which dropped.

Findings on the source

Brand and manufacturer disagreeing; rows where classification refused to guess.

Induced specs

64 category rulebooks derived from the rows, with a count behind every attribute.

Review queue

Ranked by uncertainty × sibling SKUs affected — the best use of the next human minute.

Relationship graph

614 edges: which battery powers which tool, which board varies from which.

Exports the 252-column delivery file as CSV and XLSX, plus the provenance ledger, review queue and relationship graph.

Most pipelines generate, then check. This one cannot generate.

The usual approach

Model writes the cell; a validator checks it afterwards

A confidence score invented per rule

Fill as much of the sheet as possible

Character limits checked, sometimes missed

A review queue that forgets its answers

No third-party packages · CSV and XLSX read and written with the standard library

CALIPER

Model may only propose a fact that quotes the source — composition renders, it cannot invent

Confidence rises from independent methods agreeing, and is checked against sibling consistency

Fill rate calibrated against the answer key, because over-filling is fabrication

A budget problem with abbreviation ladders — 100% compliant by construction

Corrections persist, scoped to part / family / category / brand, and replay every run

The honest edges.

A submission whose thesis is measurement has to report what it cannot measure.

- 1 Two labelled rows**

Every accuracy figure rests on a calibration set of two. The harness prints the sample size beside every rate and scales untouched to a larger file.
- 2 No manufacturer retrieval**

Finding 02 shows this, not a missing rule, is the binding constraint on attribute-value accuracy.
- 3 No official vocabularies**

The LOV, the 27,000-row manufacturer master and the UOM standards were never distributed. Our registry is a ~75-entry bootstrap, and LOV coverage is measured against an induced vocabulary and labelled as such.
- 4 Conformance is not quality**

100% character compliance is true, and is not the same as a good line. No automated check we have closes that gap.

Enrichment is table stakes.

Knowing which rows ship unattended — and being able to prove why — is the product.

PROTOTYPE

`python -m caliper serve`

REPOSITORY

`github.com/jacklachan/unihack`

WRITTEN BRIEF

`docs/submission.html`

Every number in this deck was produced by a run — then the outputs behind it were read.